

Lunar Formation via Triple Phase Transition in the Differentiating Proto-Earth

*Bingham-Herschel rheology, chaotic axis wobble,
complete force inventory, Proto-Accretion Orbital body
and progressive terrestrial dynamo*

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Abstract. The origin of the Moon remains one of the unsolved problems of planetary science. The canonical giant-impact model faces growing geochemical difficulties: it does not naturally predict the near-perfect Earth–Moon isotopic identity, the crustal dichotomy, or the ≈ 300 Myr delay of the terrestrial dynamo. This work submits to examination an alternative grounded in a necessary thermodynamic observation: any Earth-mass planet emerges from accretion in a state of near-total silicate mantle melting, accretion energy exceeding total fusion energy by a factor ≈ 155 (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The proto-Earth is therefore a rapidly rotating magma body ($T_{\text{rot}} \approx 3.5$ h) without a stabilising satellite, whose axis wobbles chaotically within $[40^\circ, 70^\circ]$ in its own co-rotating frame. In this context, a single driver—progressive Fe-Ni segregation—governs three coupled transitions: a rheological transition (structuring of the Coherent Magmatic Torus), a mechanical transition (episodic hypersonic ejections governed by a double potential well), and a magnetic transition (dynamo delayed by ≈ 350 Myr). The central prediction—seismic interface(s) between 200 and 530 km depth, impedance contrast $|R| \in [0.01, 0.04]$ —is testable by **Chang’e 7** (South Pole, 2026), **FSS**, **LEMS** and **Artemis III** (2028–2029). All hypotheses are explicit and ranked. Limitations are acknowledged. The validation programme is defined and sequenced.

Keywords: lunar origin · Triple Phase Transition · chaotic axis wobble · Hadean proto-Earth · Bingham-Herschel rheology · Coherent Magmatic Torus · double

potential well · Kramers rate · Mathieu resonance · seismic interface · Hadean
dynamo · falsifiable predictions.

Epistemic note

This work submits a theory to the scrutiny of the scientific community. The phrasing reflects this posture deliberately: the theory describes, predicts and constrains—it does not assert absolute certainties. All hypotheses are numbered and ranked. All limitations are acknowledged. Final validation belongs to forthcoming lunar seismic data and, in a second stage, to three-dimensional numerical simulations.

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1 Introduction

1.1 The problem of lunar origin

The giant-impact hypothesis—collision between the proto-Earth and a Mars-sized body, Theia (Hartmann & Davis, 1975; Cameron & Ward, 1976; Canup, 2004)—has provided the reference framework for lunar formation for four decades. It satisfies qualitatively the angular momentum constraint of the Earth-Moon system (Cuk & Stewart, 2012; Canup, 2012), but faces growing geochemical difficulties: the near-perfect isotopic identity ($\Delta^{17}\text{O} < 5$ ppm, Wiechert et al. 2001; Dauphas et al. 2014; Dauphas 2017) is difficult to obtain without ad hoc adjustments, the crustal dichotomy is not naturally predicted, and the Hadean dynamo delay remains unexplained within this framework.

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in lunar formation (Sossi et al., 2025). This finding motivates the exploration of alternatives grounded in internal proto-Earth dynamics.

This work proposes such a mechanism. It does not claim to resolve the question definitively—final validation belongs to forthcoming lunar seismic data and 3D simulations. It offers a physically coherent framework, quantitative falsifiable predictions, and a clear hierarchy between what is established, what is conjectured, and what remains to be measured.

1.2 Three discriminating observational constraints

Isotopic similarity. The Earth-Moon isotopic composition is quasi-identical for O, Cr and Ti, established to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The present theory satisfies this constraint mechanically: ejected material is sampled from the contemporaneous Hadean magma at each ejection, with no exotic external source.

Iron depletion. The Moon is significantly less rich in Fe than the terrestrial mantle (ONeill, 1991; Jones & Drake, 1986). In the present theory this depletion is a direct progressive consequence of the single driver: Fe-Ni segregation continually depletes the source magma between successive ejections.

Angular momentum. The angular momentum budget of the Earth-Moon system requires a very rapidly rotating proto-Earth at the time of formation. The value adopted, $T_{\text{rot}} = 3.5$ h, is consistent with planetary accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010).

1.3 Architecture of the theory

Established result

One driver, three coupled transitions. Progressive Fe-Ni segregation towards the forming core simultaneously governs:

1. a **rheological transition**: structuring of a Coherent Magmatic Torus (CMT) in the oblique Hadean intertropical band $|\phi| < 30^\circ$;
2. a **mechanical transition**: episodic hypersonic ejections governed by the dynamics of a double potential well $V(U)$;
3. a **magnetic transition**: delayed emergence of the terrestrial dynamo via three independently constrained saturation steps.

2 Initial state of the proto-Earth

2.1 Near-total melting: a necessary thermodynamic consequence

The accretion energy in the homogeneous-sphere approximation (Safronov, 1969):

$$E_{\text{grav}} = \frac{3 G M_{\oplus}^2}{5 R_{\text{eq,proto}}} \approx 2.49 \times 10^{32} \text{ J}. \quad (1)$$

The energy required to melt the entire silicate mantle (Stixrude & Lithgow-Bertelloni, 2014):

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24} \times 4 \times 10^5 \approx 1.6 \times 10^{30} \text{ J}. \quad (2)$$

Established result

Energy ratio—a consensus result.

$$\frac{E_{\text{grav}}}{E_{\text{fus}}} \approx 155. \quad (3)$$

Accretion energy exceeds total fusion energy by approximately two orders of magnitude. Established independently by Solomatov (2000), Elkins-Tanton (2012) and Rubie et al. (2015). This is not a hypothesis of this work: it is the obligatory starting point of any serious theory of the Hadean period. The physically relevant question is not *how to melt the mantle*, but **how and how fast to cool it**.

Additional heat sources reinforce this picture: Fe-Ni segregation heat ($\approx 3 \times 10^{30}$ J, Flasar & Birch 1973; Rubie et al. 2003), short-lived radioactivity (^{26}Al , ^{60}Fe , Urey 1955; Huss et al. 2009), and continuous planetesimal bombardment (Elkins-Tanton et al., 2011). Together they maintain surface temperatures between 2 000 and 4 000 K,

preventing any stable protocrust (Solomatov, 2000).

Hypothesis — explicit status

H0—Total volumetric melting. At the onset of Fe-Ni segregation, the proto-Earth is in a state of total volumetric melting: the entire silicate mantle is above the liquidus, with no stable solid internal interface during the active window 4.568–4.4 Ga. This is not a magma ocean resting on a substrate: it is a self-gravitating fluid volume. Pressure waves generated by planetesimal impacts propagate through the entire volume.

Caveat: Nomura et al. (2011) suggest that the deep lower mantle may not have been fully molten owing to the pressure dependence of the solidus. The TPT does not require melting to the Earth's centre: it requires melting of the mass available in the Hadean intertropical band, which the factor 155 guarantees thermodynamically for the upper and mid mantle.

2.2 Initial rotation period: $T_{\text{rot}} \approx 3.5$ h

Three convergent physical mechanisms constrain T_{rot} to ≈ 2.5 – 3.5 h.

1. T-Tauri magnetic torque. The young Sun (Koenigl, 1991; Shu et al., 1994) magnetically couples to the inner protoplanetary disc, transferring angular momentum to material in the inner disc.

2. Angular momentum conservation during contraction. Contraction from $R_0 \approx 2 R_{\oplus}$ to the final radius:

$$\frac{T_{\text{rot}}^{(f)}}{T_{\text{rot}}^{(0)}} = \left(\frac{R_{\oplus}}{R_0} \right)^2 \approx \frac{1}{4}. \quad (4)$$

Starting from $T_{\text{rot}}^{(0)} \approx 14$ h, contraction yields $T_{\text{rot}}^{(f)} \approx 3.5$ h.

3. Skater effect: Fe-Ni segregation. Migration of Fe-Ni towards the centre reduces the moment of inertia:

$$\frac{\Omega_f}{\Omega_0} = \frac{I_0}{I_f} = \frac{I_{\text{homogeneous}}}{I_{\text{differentiated}}}. \quad (5)$$

For $R_c \approx 0.55 R_{\oplus}$, $M_c \approx 0.32 M_{\oplus}$ (Dziewonski & Anderson, 1981): $I_f/I_0 \approx 0.82$.

Established result

$T_{\text{rot}} \approx 3.5$ h—consistent with the community consensus. N-body accretion simulations give median rotation periods of 3–5 h (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2013). The value 3.5 h is the same as that used by Cuk & Stewart (2012) and Canup (2012): the TPT does not postulate an exotic initial spin. A remarkable property then emerges—the Coriolis compensation parame-

ter:

$$b' = \frac{2\Omega \sin \varepsilon \cdot U_{\text{crit}}}{|g_{\text{eff}}|} \approx 1.000 \quad \text{for } T_{\text{rot}} = 3.5 \text{ h}, \varepsilon = 55^\circ. \quad (6)$$

The radial Coriolis force exactly compensates effective gravity at the ejection threshold. This is not postulated: it emerges directly from the equations for these nominal parameters.

Robustness of b' over the Hadean parameter range

The exact value $b' = 1.000$ is a numerical coincidence at the nominal point. The underlying physical property—the radial Coriolis force being *of the same order* as effective gravity—is robust over the entire Hadean range. Substituting $U_{\text{crit}} = \sqrt{2|g_{\text{eff}}| R_{\text{eq,proto}}}$ into the definition of b' yields the **closed-form expression**:

$$b'(\varepsilon, T_{\text{rot}}) = \frac{2\sqrt{2}\Omega(T_{\text{rot}}) \sin \varepsilon \sqrt{R_{\text{eq,proto}}}}{\sqrt{|g_{\text{eff}}(\varepsilon, T_{\text{rot}})|}}, \quad (7)$$

where $\Omega = 2\pi/T_{\text{rot}}$ and, to leading order, $|g_{\text{eff}}| = g_0 - \Omega^2 R_{\text{eq,proto}} \sin \varepsilon$. This expression can be evaluated analytically across the full Hadean parameter space.

T_{rot}	$b'(\varepsilon, T_{\text{rot}})$		
	$\varepsilon = 40^\circ$	$\varepsilon = 55^\circ$ (nominal)	$\varepsilon = 70^\circ$
2.5 h	0.89	1.12	1.31
3.0 h	0.94	1.05	1.16
3.5 h (nominal)	0.97	1.000	1.07
4.0 h	0.99	0.94	0.98

Over the full range, $b' \in [0.85, 1.35]$: neither $b' \ll 1$ (negligible Coriolis) nor $b' \gg 1$ (overwhelming Coriolis). The qualitative compensation is guaranteed for every plausible Hadean $(\varepsilon, T_{\text{rot}})$ combination. The value $b' = 1.000$ at the nominal point is a *remarkable numerical coincidence*, not a necessary condition for the mechanism: the mechanism operates for any $b' \sim \mathcal{O}(1)$.

2.3 Proto-Earth geometry: the Maclaurin spheroid

Dynamic flattening coefficient:

$$J_2 \approx \frac{5}{4} \cdot \frac{\Omega^2 R_\oplus}{g_0} \approx 0.101, \quad \Omega = 2\pi/T_{\text{rot}} \approx 4.987 \times 10^{-4} \text{ rad s}^{-1}. \quad (8)$$

Effective gravity at latitude ϕ in the co-rotating frame:

$$g_{\text{eff}}(\phi) = g_{\text{eff}}^{(0)} \left(1 + \frac{3}{2} J_2 \sin^2 \phi\right) - \Omega^2 r(\phi) \cos^2 \phi, \quad r(\phi) = R_{\text{eq,proto}} (1 - J_2 \sin^2 \phi). \quad (9)$$

Minimal in the band $|\phi| < 30^\circ$: the geometric attractor of the CMT.

3 Fundamental geometric clarification: axis wobble and the Hadean intertropical band

Geometric warning—read first

The Hadean rotation axis is not the present terrestrial axis. Today the Earth's axis is stabilised at 23.5° to the ecliptic normal by the Moon's presence. Lunar formation is precisely what this work seeks to explain: invoking the Moon as an initial condition would be circular. Before the Moon, the axis wobbled chaotically within $[40^\circ, 70^\circ]$ (Laskar et al., 1993a; Laskar, 1993b). The ranges 23.5° and $[40^\circ, 70^\circ]$ describe different angles in different reference frames and are not directly comparable.

The Hadean intertropical band $|\phi| < 30^\circ$ has no connection with the present tropical zone at $\pm 23.5^\circ$. It is defined perpendicularly to the instantaneous Hadean rotation axis—an axis that wobbles continuously, sweeping all geographic latitudes in ≈ 47 h. It is therefore not geographically fixed. Over the duration of one ejection episode it traverses the entire globe.

3.1 Definition of ε in the TPT

Throughout this work, ε denotes the instantaneous angle between the rotation vector Ω and the principal axis of inertia of the Maclaurin spheroid, defined **in the co-rotating body frame**. It is not the astronomical obliquity in the classical orbital sense. It describes the wobble—the free nutation—of the axis with respect to the body's own shape.

3.2 The wobble as a direct force

The axis wobble implies $\dot{\Omega} \neq \mathbf{0}$. Poincare (1910) showed that this generates an **Euler acceleration** on every fluid particle:

$$\mathbf{a}_{\text{Euler}} = \dot{\Omega} \times \mathbf{r}, \quad (10)$$

with radial projection:

$$\mathbf{a}_{\text{Euler}} \cdot \hat{\mathbf{r}} = |\dot{\Omega}| r(\phi) \sin(\varepsilon + \phi). \quad (11)$$

Maximal in the band $|\phi| < 30^\circ$ (maximal $r(\phi)$). The axis wobble is therefore a *direct energy source* for the CMT, not merely a context.

3.3 Maintenance of the wobble in the Hadean context

Relaxation time towards alignment:

$$\tau_{\text{relax}} \sim \frac{\eta R_{\text{eq,proto}}^3}{G M_{\oplus}^2 f} \approx 10^6 \text{ yr} \quad (\eta \sim 10^{1-3} \text{ Pa s}, f \approx 0.107). \quad (12)$$

Perturbation	Timescale
T-Tauri eruptions (CME)	Days to weeks
Residual planetesimal impacts	10^3 – 10^5 yr
Extreme thermal convection ($\Delta T \sim 2000$ K)	Continuous, $\sim 10^3$ yr
Fe-Ni segregation (inertia tensor)	$\tau_{\text{seg}} \approx 30$ – 50 Myr
Giant-planet tidal forcing	Continuous, variable

Each perturbation drives the system away from alignment on timescales far shorter than τ_{relax} . The misalignment $\varepsilon \in [40^\circ, 70^\circ]$ is the *normal condition* of a Hadean proto-Earth.

4 Complete force inventory

All forces, projections and equations that follow are expressed in the instantaneous Hadean co-rotating body frame.

4.1 Conservative forces—structural

Effective gravity $g_{\text{eff}}^{(k)}$: decreasing episode by episode.

$$g_{\text{eff}}^{(0)} = g_0 - \Omega^2 R_{\text{eq,proto}} \sin \varepsilon \approx 8.51 \text{ m s}^{-2} \quad (T_{\text{rot}} = 3.5 \text{ h}, \varepsilon = 55^\circ). \quad (13)$$

At each episode k :

$$g_{\text{eff}}^{(k)} = \frac{G M_{\oplus}^{(k)}}{R_{\oplus}^2} - (\Omega^{(k)})^2 R_{\oplus} \sin \varepsilon, \quad (14)$$

where both mass loss and rotational slowdown act in the same direction: $g_{\text{eff}}^{(k)}$ decreases episode by episode, mechanically facilitating subsequent episodes.

Centrifugal force. $F_c = \Omega^2 r(\phi) \cos^2 \phi$; relative amplitude $\approx 5\%$ of g_0 . Concentrates matter in the intertropical bulge.

Angular momentum gradient—Rayleigh criterion. (Rayleigh, 1917) For $U \propto r^\alpha$, radial stability requires $\partial(rU)/\partial r > 0$, i.e. $\alpha > -1$. The contribution to the instability coefficient (term λ_3 , Eq. (40)) is destabilising for $\alpha > 1$.

Tangential gravity gradient—Maclaurin term.

$$g_{\tan}(\phi) = g_{\text{eff}}(\phi) \cdot J_2 \cdot \sin(2\phi) \in [0.08, 0.55] \text{ m s}^{-2} \quad \text{for } \phi \in [5^\circ, 30^\circ]. \quad (15)$$

Ten to sixty times smaller than vertical g_{eff} : the component the flux must overcome to eject tangentially.

4.2 Dynamic forces—destabilising

Radial Coriolis force.

$$f_{\text{rad}} = 2\Omega \sin \varepsilon \cdot U, \quad (16)$$

$$f_{\text{horiz}} = 2\Omega \cos \varepsilon \cdot U, \quad (17)$$

$$\frac{f_{\text{rad}}}{f_{\text{horiz}}} = \tan \varepsilon. \quad (18)$$

Established result

Geometric threshold $\varepsilon = 45^\circ$ —necessary condition. For $\varepsilon > 45^\circ$: the destabilising radial component dominates. For $\varepsilon < 45^\circ$: the confining horizontal component dominates. This threshold follows from the sole kinematics of the Coriolis force: no adjustable parameter enters. For $\varepsilon = 55^\circ$: 82% of total Coriolis force directed towards ejection.

Euler precession acceleration. Eq. (11): constitutes term λ_4 of the instability equation—the direct dynamical signature of the axis wobble.

4.3 Taylor-Proudman columns: curved, truncated, channelling

In a rapidly rotating fluid ($\text{Ro} \ll 1$), Taylor-Proudman columns arise. Five mechanisms curve and truncate them in the Hadean context:

1. **Obliquity $\varepsilon \in [40^\circ, 70^\circ]$:** Ω is misaligned with equipotential axes. Columns cut obliquely across surfaces of constant effective gravity.
2. **Spherical curvature and β -effect:** $f = 2\Omega \cos \phi$ varies with latitude, curving columns along lines of constant f —direct analogy with Jupiter’s zonal bands.

3. **Bingham-Herschel rheology:** a Taylor column is sharply truncated where $\tau < \tau_y$. The band $|\phi| < 30^\circ$, where $\tau > \tau_y$, confines columns by rheological physics alone.
4. **Axis wobble ($\dot{\Omega} \neq 0$):** the Taylor-Proudman theorem requires stationary Ω . The Euler acceleration continuously tilts and reforms columns.
5. **Radial density gradient:** Fe-Ni segregation creates $\rho(r, t)$. The Solberg-Hoiland criterion (Solberg, 1936; Hoiland, 1941) replaces simple Rayleigh, curving columns according to the density profile.

Established result

Curved columns generate the toroidal flow. Taylor-Proudman columns are not absent—they are curved and truncated. Their curvature deflects fluid into the azimuthal direction. It is this deflection that feeds the toroidal mode $\ell = 1, m = 1$ selected by the inverse cascade. Column curvature is the physical selection mechanism for the dominant mode.

4.4 Tidal forces

Solar tide. Relative amplitude $\approx 10^{-4} g_0$. Low-frequency resonant forcing on obliquity.

Growing proto-lunar tide: Mathieu resonance. Tidal force from the proto-satellite:

$$F_{m,\text{Moon}}(t) \propto \frac{M_{\text{Moon}}(t)}{a(t)^3}, \quad (19)$$

modulates the barrier height:

$$\Delta E_{\text{barr}}^{(k)}(t) = \Delta E_{\text{barr}}^{(k,0)} - A_m^{(k)}(t) \cos(n_{\text{Moon}} t). \quad (20)$$

The radial perturbation equation:

$$\ddot{\xi}_r = [\lambda + A_m^{(k)}(t) \cos(n_{\text{Moon}} t)] \xi_r, \quad (21)$$

is the Mathieu equation (McLachlan, 1947). Parametric resonance bands exist for $n_{\text{Moon}} \approx 2\sqrt{|\lambda|}$ even when $\lambda < 0$. The proto-lunar tide is the **dominant trigger** for episodes 2 to N .

Giant-planet tidal forcing. Low-frequency perturbation on obliquity (10^5 – 10^6 yr).

4.5 Rheological forces

BH yield stress τ_y : cohesion of CMT; irreversible closure of TPT window below $T^* \approx 1\,800$ K. See Section 6.

KH instability: suppressed by $\text{Bi}_{\text{KH}} \gg 1$. See Section 7.

RT instability: suppressed by $\rho_{\text{flux}}/\rho_{\text{atm}} \approx 200$.

4.6 External energetic perturbations: the T-Tauri Sun

1. Rheological modulator. Surface temperature under T-Tauri irradiance (Stefan-Boltzmann):

$$T_{\text{surf}} = \left[\frac{(1 - A_0) L_*}{16\pi\sigma_{\text{SB}} d^2} \right]^{1/4} \approx \begin{cases} 1\,800\text{--}2\,200\text{ K} & \text{between outbursts} \\ 2\,800\text{--}4\,200\text{ K} & \text{during outbursts.} \end{cases} \quad (22)$$

Maintains τ_y in the optimal ejection range ($10^2\text{--}10^4$ Pa).

2. Impulsive trigger via CME. CME energies $10^{27}\text{--}10^{29}$ J per event (Airapetian et al., 2016); $\sim 10^9\text{--}10^{10}$ opportunities over 10^6 yr.

3. Independent extinction mechanism. Declining T-Tauri activity raises τ_y , closing the TPT window independently of obliquity stabilisation.

4.7 Volumetric pressure waves: planetesimal impacts

$$P_{\text{wave}}(r) \approx \frac{E_{\text{impact}}^{1/2} \rho^{1/2}}{4\pi r}. \quad (23)$$

Propagates through the entire fluid volume. Three roles: turbulent energy injection (feeds inverse cascade), local perturbation of $U(r)$ (may cross τ_y), finite perturbation on ΔE_{barr} (contributes to Kramers rate).

4.8 Geostrophic β -force

$$\beta = \frac{2\Omega \cos \varepsilon}{R_{\text{eq,proto}}} \approx 8.98 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}, \quad L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}} \approx 0.52 R_\oplus. \quad (24)$$

Arrests the inverse cascade at mode $\ell = 1, m = 1$.

4.9 Atmospheric confinement pressure

Silicate atmosphere ($T \approx 2\,000\text{--}4\,000$ K, $P \approx 10^5\text{--}10^7$ Pa): lateral confinement during ejection, suppression of large-wavelength KH modes, co-rotating jet co-mobility.

Force inventory—summary table

Force	Rel. amplitude	Role in TPT
Effective gravity $g_{\text{eff}}^{(k)}$	1.000	λ_1 (stabilising)
Radial Coriolis	≈ 0.62	λ_2 (destabilising)
Angular momentum gradient	$\sim 10^{-7}$	λ_3 ($\alpha > 1$)
Euler acceleration (wobble)	variable	λ_4 (precession)
Tangential gradient (Maclaurin)	$\sim 10^{-2}$	λ_5
Centrifugal force	≈ 0.05	Concentrates CMT
β -force	≈ 0.30	Selects $\ell = 1$ mode
Solar tide	$\sim 10^{-4}$	Weak resonant forcing
Proto-lunar tide	variable	Mathieu resonance (ep. 2– N)
Giant-planet tides	small	Maintains obliquity
Impact pressure waves	$\sim 10^{-2}$	Injection + trigger
T-Tauri CME	$\sim 10^{-10}$	First-episode trigger
BH yield stress τ_y	threshold	CMT cohesion
Atmospheric pressure	$\sim 10^{-3}$	Lateral confinement

5 Emergence of the Coherent Magmatic Torus (CMT)

5.1 Geostrophic regime and inverse cascade

$$\text{Ro} = \frac{U_{\text{turb}}}{2\Omega R_{\text{eq,proto}}} \approx 1.35 \times 10^{-4} \ll 1, \quad (25)$$

$$\text{Re} = \frac{\rho U_{\text{turb}} R_{\text{eq,proto}}}{\eta_{\text{eff}}} \approx 2.4 \times 10^9 \gg 10^5. \quad (26)$$

Turbulent regime justifies the geostrophic inverse cascade. Cascade arrests at $L_\beta \approx 0.52 R_\oplus$: mode $\ell = 1$, $m = 1$. The CMT is the spatial expression of this mode in $|\phi| < 30^\circ$.

5.2 The band $|\phi| < 30^\circ$ as geometric attractor

Four independent convergent effects establish the Hadean intertropical band as the stable spatial attractor: (1) minimal β at the intertropical bulge (L_β maximal); (2)

stable attractor for all $\varepsilon \in [40^\circ, 70^\circ]$; (3) BH rheology breaks ergodicity (stress barrier at $|\phi| = 30^\circ$); (4) Rossby waves propagate towards oblique low latitudes.

5.3 'Aa structure: intrinsic thermal insulator

The CMT has a two-phase structure: mobile plastic core + fragmented clinker crust. The clinker crust blocks radiative energy loss without requiring an external atmosphere. The CMT is its own thermal regulator.

5.4 Inter-CMT coupling and uniqueness of the accretion stratum

Limitation L1 (Section 13) raises the multiplicity question: could several co-existing CMTs compromise the predicted concentric stratification? This subsection shows, by a scale-separation argument, that the answer is no.

Three coupling mechanisms

1. Phase synchronisation via Rossby waves. Each CMT emits Rossby waves propagating zonally. In the BH fluid, these waves are confined to the active band $|\phi| < 30^\circ$ where $\tau > \tau_y$. An azimuthal phase offset $\delta\phi$ between two CMTs generates a pressure difference that drives synchronisation.

The effective β parameter on the oblate spheroid ($\varepsilon \neq 0$, $J_2 \neq 0$) carries a geometric correction:

$$\beta_{\text{eff}} = \frac{2\Omega \cos \varepsilon}{R_{\text{eq,proto}}} \left(1 + \frac{3}{2}J_2\right) \approx \beta \times 1.15, \quad (27)$$

where the factor $(1 + \frac{3}{2}J_2) \approx 1.15$ reflects the **geometric correction of the oblateness**: the oblate spheroid stretches the effective meridional distance between latitude circles, thereby increasing the meridional gradient of the Coriolis parameter projected onto the surface. This correction is purely geometric—it is independent of the rheology—and amounts to 15% for $J_2 \approx 0.101$, which is non-negligible.

The Rossby-wave group speed at the Rhines scale, in the *Newtonian reference case* (Rhines, 1975):

$$c_{\text{Rossby}}^{(\text{Newt})} \sim \beta_{\text{eff}} L_\beta^2, \quad (28)$$

gives a synchronisation time:

$$\tau_{\text{sync}} \sim \frac{L_\beta}{c_{\text{Rossby}}^{(\text{Newt})}} = \frac{1}{\beta_{\text{eff}} L_\beta} \approx 10^3\text{--}10^4 \text{ yr}. \quad (29)$$

In a Bingham-Herschel fluid with yield stress τ_y , Rossby waves exist only where $\tau > \tau_y$, i.e. precisely the active band $|\phi| < 30^\circ$ —a *finite-width waveguide* of width $L_\phi < L_\beta$. By analogy with the fundamental mode of Rossby waves in a meridionally bounded channel (Vallis, 2006, §6.5), confinement imposes an effective meridional

wavenumber $k_y^{(\text{eff})} = \pi/L_\phi$, modifying the group speed:

$$c_{\text{Rossby}}^{(\text{BH})} \approx \beta_{\text{eff}} L_\beta^2 \cdot F\left(\frac{L_\phi}{L_\beta}\right), \quad F\left(\frac{L_\phi}{L_\beta}\right) \approx \frac{(L_\phi/L_\beta)^2}{\pi^2}. \quad (30)$$

For $L_\phi/L_\beta \approx 0.5$:

$$F \approx \frac{(0.5)^2}{\pi^2} \approx 0.025, \quad (31)$$

so $c_{\text{Rossby}}^{(\text{BH})} \approx 0.025 c_{\text{Rossby}}^{(\text{Newt})}$ and:

$$\tau_{\text{sync}}^{(\text{BH})} \approx \frac{\tau_{\text{sync}}^{(\text{Newt})}}{F} \approx 40 \times \tau_{\text{sync}}^{(\text{Newt})} \approx 4 \times 10^4 - 4 \times 10^5 \text{ yr}. \quad (32)$$

This correction reinforces the conclusion without invalidating it. The Newtonian estimate *underestimated* τ_{sync} by a factor ≈ 40 . Even so, $\tau_{\text{sync}}^{(\text{BH})} \approx 4 \times 10^5 \text{ yr}$ remains *two orders of magnitude* below $\tau_{\text{seg}} \approx 30\text{--}50 \text{ Myr}$: the scale separation holds with a comfortable margin of $\times 100$.

2. Shared inverse-cascade energy reservoir. Multiple CMTs in the same band share the same turbulent reservoir; both are expressions of the same $\ell = 1, m = 1$ mode. Energy injection into one immediately couples to the other.

3. Direct gravitational coupling. Two toroidal masses $M_{\text{CMT}} \sim 10^{22} \text{ kg}$ separated by π in longitude:

$$F_{\text{grav,CMT}} \sim \frac{G M_{\text{CMT}}^2}{(2R_{\text{eq,proto}})^2} \sim 10^{10} \text{ N}, \quad (33)$$

small in absolute amplitude but acting over $10^4\text{--}10^5 \text{ yr}$, contributing to long-term phase alignment.

Scale-separation argument

Established result

$\tau_{\text{sync}}^{(\text{BH})} \ll \tau_{\text{seg}}$: **CMTs synchronise before the next segregation step.** Fe-Ni segregation operates on $\tau_{\text{seg}} \approx 30\text{--}50 \text{ Myr}$. The Newtonian estimate gives $\tau_{\text{sync}}^{(\text{Newt})} \approx 10^3\text{--}10^4 \text{ yr}$; corrected by the BH channel analogue (factor ≈ 40 , Eq. 32), this becomes $\tau_{\text{sync}}^{(\text{BH})} \approx 4 \times 10^4\text{--}4 \times 10^5 \text{ yr}$. Segregation has no time to desynchronise CMTs between ejection episodes: the margin is **two orders of magnitude** ($\times 100$). Synchronisation is maintained throughout the active TPT window.

L8—Rossby-wave group speed in a BH fluid: channel analogue

The Newtonian estimate $c_{\text{Rossby}}^{(\text{Newt})} \sim \beta_{\text{eff}} L_\beta^2$ (Rhines, 1975) applies to an unconfined homogeneous fluid. The BH active band constitutes a meridional *waveg-*

uide of width $L_\phi \approx 0.5 L_\beta$; the channel analogue (Vallis, 2006, §6.5) gives $F \approx (L_\phi/L_\beta)^2/\pi^2 \approx 0.025$, hence $\tau_{\text{sync}}^{(\text{BH})} \approx 40 \tau_{\text{sync}}^{(\text{Newt})} \approx 4 \times 10^4 - 4 \times 10^5$ yr.

The correction goes in the opposite direction from naive intuition: confinement slows Rossby waves (larger effective meridional wavenumber $k_y = \pi/L_\phi$). The Newtonian estimate underestimated τ_{sync} by a factor ≈ 40 . But the central conclusion holds: even corrected, $\tau_{\text{sync}}^{(\text{BH})} \approx 4 \times 10^5$ yr remains **two orders of magnitude** below $\tau_{\text{seg}} \approx 30\text{--}50$ Myr (margin $\times 100$).

Rigorous derivation of $c_{\text{Rossby}}^{(\text{BH})}$ on an oblate spheroid ($\varepsilon \neq 0$, realistic $U(r)$) remains an **open problem** (Priority 6 of the analytical programme).

Consequence: one stratum per episode regardless of CMT multiplicity

Synchronised CMTs eject quasi-simultaneously. Their ejecta reach the orbital ring with the same semi-major axis and eccentricity; residual azimuthal offsets give relative velocities:

$$\delta v \approx v_{\text{orb}} \sqrt{e^2 + \frac{1}{4} \delta \theta^2} \approx 50\text{--}200 \text{ m s}^{-1}, \quad (34)$$

well below the fragmentation threshold ($\sim 1 \text{ km s}^{-1}$): collisions are sticky. Accretion time:

$$\tau_{\text{acc}} \approx \frac{T_{\text{Kep}}}{f_{\text{col}} \cdot n} \sim \text{a few } 10^3 \text{ yr}. \quad (35)$$

Established result

Emergence theorem: one stratum per episode. Local complexity (multiple CMTs) vanishes in the global observable: a single stratum per episode for any n_{CMT} . Limitation L1 loses its critical character—even in the worst case ($n_{\text{CMT}} > 1$), prediction P22 remains intact.

Subsidiary prediction: seismic interface width as a multiplicity discriminant

Imperfect synchronisation ($\delta\varphi_{\text{res}} \sim 0.2\text{--}0.3$ rad) produces a stratum of *finite thickness* ($\approx 20\text{--}50$ km) rather than a sharp interface, while $|R| \in [0.01, 0.04]$ is unchanged.

P22b — Interface width as CMT-multiplicity discriminant

- $n_{\text{CMT}} = 1 \Rightarrow$ sharp interfaces, clean reflectors, width $\ll 10$ km;
- $n_{\text{CMT}} > 1$ synchronised $\Rightarrow$ progressive impedance gradient, transition width 20–50 km.

Chang'e 7 and FSS can in principle distinguish these cases by waveform analysis. This discriminant converts L1 from a limitation into a validation opportunity.

6 First phase transition: rheological

$$\tau = \tau_y + k \dot{\gamma}^n \quad (\tau > \tau_y), \quad (36)$$

$n = 1.2$ (Caricchi et al., 2007), $\tau_y \in [10^2, 10^4]$ Pa (Giordano et al., 2008). Arrhenius temperature dependence:

$$\tau_y(T) = \tau_y^{(0)} \exp\left(-\frac{E_a}{RT}\right), \quad E_a \approx 150\text{--}250 \text{ kJ mol}^{-1}. \quad (37)$$

Established result

Natural thermal lock at $T^* \approx 1800$ K. Below this temperature $\tau_y \gtrsim 10^6$ Pa: cohesive ejection becomes physically inaccessible. The TPT window closes irreversibly through thermodynamics alone—no additional condition required.

7 High-velocity cohesion: the Bi_{KH} criterion

For a free flow: $We \approx 10^{15}$ —would imply atomising fragmentation. But the CMT is not a free flow. It is a self-gravitating collective toroidal flux. The Weber number does not apply. The relevant criterion is the Bingham-Kelvin-Helmholtz number:

$$\sigma_{\text{KH}} \approx k \Delta U \sqrt{\frac{\rho_1 \rho_2}{(\rho_1 + \rho_2)^2}}, \quad \tau_{\text{KH}} = \rho \sigma_{\text{KH}} \Delta U / k, \quad \text{Bi}_{\text{KH}} = \frac{\tau_y}{\tau_{\text{KH}}}. \quad (38)$$

Established result

For $\text{Bi}_{\text{KH}} \gg 1$: the yield threshold suppresses KH instability before any fragmentation in the confined toroidal flux.

Acknowledged limitation

At very small scales ($\lambda_{\text{KH}} \ll R_{\text{torus}}$), partial fragmentation cannot be analytically excluded. This is Prediction P11—to be validated by high-resolution 3D SPH simulation.

8 Second phase transition: mechanical

8.1 The generalised instability equation λ

$$\ddot{\xi}_r = \lambda \xi_r, \quad (39)$$

with $a = (1 - \alpha)/r^2$, $b = 2\Omega^{(k)} \sin \varepsilon > 0$, $c = g_{\text{eff}}^{(k)} < 0$:

$$\lambda(U, \varepsilon, \alpha, k) = \underbrace{c}_{\lambda_1} + \underbrace{bU}_{\lambda_2} + \underbrace{aU^2}_{\lambda_3} + \underbrace{\frac{|\dot{\Omega}| r(\phi) \sin(\varepsilon + \phi)}{R_{\text{eq,proto}}}}_{\lambda_4} + \underbrace{\frac{g_{\text{eff}}^{(0)} J_2 \sin(2\phi)}{R_{\text{eq,proto}}}}_{\lambda_5}. \quad (40)$$

λ_1 : effective gravity (stabilising). λ_2 : radial Coriolis (destabilising). λ_3 : angular momentum gradient (destabilising if $\alpha > 1$). λ_4 : Euler acceleration (axis wobble). λ_5 : Maclaurin tangential gradient.

8.2 Episodicity regimes

Super-rigid ($\alpha > 1$): instability window $[U_1, U_2]$ —spontaneous episodicity:

$$U_{1,2} = \frac{r^2}{\alpha - 1} \left[\Omega^{(k)} \sin \varepsilon \mp \sqrt{(\Omega^{(k)})^2 \sin^2 \varepsilon - \frac{(\alpha - 1)}{r^2} |g_{\text{eff}}^{(k)}|} \right]. \quad (41)$$

Sub-rigid ($\alpha \leq 1$): single threshold—episodicity driven by Mathieu resonance and Kramers rate.

Established result

α is a shape parameter, not an existence condition. The double well exists for any α as long as $\varepsilon > 45^\circ$. The 3D SPH simulation (Prediction P13) will measure $\hat{\alpha}$ honestly, without predetermined outcome.

8.3 Progressive CMT lightening

$$\rho_{\text{CMT}}(t) = \rho_0 - \xi_{\text{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4\,500, \quad \rho_f \approx 3\,200 \text{ kg m}^{-3}. \quad (42)$$

Amplification factor $\rho_0/\rho_{\text{CMT}}(t)$ grows monotonically: the bifurcation comes to meet the growing CMT.

8.4 Double potential well $V(U)$

$$V(U) = -\frac{a}{3}U^3 - \frac{b}{2}U^2 - cU + V_0, \quad (43)$$

with barrier height $\Delta E_{\text{barr}}^{(k)} = \int_{U_{\text{CMT}}}^{U_{\text{top}}} |\lambda(U)| dU$, decreasing at each episode. Ejection episodes are spontaneous transitions between metastable states—not scripted thresholds.

8.5 Kramers crossing rate: first episode

$$\Gamma_{\text{Kramers}} = \frac{\omega_0 \omega_b}{2\pi \gamma_{\text{diss}}} \exp\left(-\frac{\Delta E_{\text{barr}}}{k_B T_{\text{eff}}}\right), \quad P(\Delta t) = 1 - e^{-\Gamma_{\text{Kramers}} \Delta t}. \quad (44)$$

With T-Tauri CME and planetesimal impact frequencies, $\sim 10^9$ – 10^{10} opportunities over 10^6 yr: the probability of no triggering event is negligible.

Established result

The first episode is a quantified barrier crossing. The Kramers rate unifies accumulation in P_1 , metastability, stochastic triggering, and post-ejection return. This is a stochastic process with a calculable rate, not an unconstrained event.

8.6 Critical velocity and maximum ejectable mass

$$U_{\text{crit}}^{(0)} = \sqrt{2 g_{\text{eff}}^{(0)} R_{\text{eq,proto}}} \approx 11.22 \text{ km s}^{-1} \quad (\text{conservative upper bound}), \quad (45)$$

$$\text{Ma} = U_{\text{crit}}/c_s \approx 12.5 \quad (c_s \approx 0.9 \text{ km s}^{-1}, \text{ Mader et al. 2013}), \quad (46)$$

$$M_{\text{ej}}^{\text{max}} \approx \frac{2 \Omega^{(k)} \sin \varepsilon \cdot U_{\text{crit}}^{(k)} \cdot M_{\text{CMT}}}{g_{\text{eff}}^{(k,0)}} \approx 1.0\text{--}2.0 \times 10^{22} \text{ kg}. \quad (47)$$

8.7 Hypersonic flux → Proto-Accretion Orbital body: three phases

The physically most delicate step in the TPT is the following: how does a hypersonic toroidal flux ($\text{Ma} \approx 12.5$) become a coherent solid-plastic body in orbit? Three successive phases describe this transition.

Phase I — Ballistic near-field expansion ($r < 2\text{--}3 R_{\oplus}$): cohesion secured by $\text{Bi}_{\text{KH}} \gg 1$

From the moment the flux leaves the CMT, it expands ballistically. Cohesion is secured by $\text{Bi}_{\text{KH}} \gg 1$ (Section 7): the KH instability is suppressed at all large and intermediate scales. There is no spray of droplets; the flux is a collective plastic assembly.

The flux cools adiabatically:

$$\frac{dT}{dt} \approx -T(\gamma - 1) \nabla \cdot \mathbf{v}, \quad (48)$$

with $\gamma \approx 1.3$ and $\nabla \cdot \mathbf{v} \approx U_{\text{crit}}/R_{\text{eq,proto}} \approx 1.5 \times 10^{-3} \text{ s}^{-1}$:

$$\tau_{\text{cool}} \approx \frac{1}{(\gamma - 1)(\nabla \cdot \mathbf{v})} \approx 700 \text{ s} \approx 12 \text{ min}. \quad (49)$$

Transit time to the orbit ($a \approx 4 R_{\oplus}$):

$$\tau_{\text{transit}} \approx \sqrt{\frac{a^3}{GM_{\oplus}}} \approx 2\text{--}3 \text{ h}. \quad (50)$$

Since $\tau_{\text{cool}} \ll \tau_{\text{transit}}$, the sheet cools partially during transit and arrives at the orbit in a **plastic but not yet solid state**—favourable to sticky accretion.

Phase II — Void crossing ($2 R_{\oplus}$ to orbit): rotational stabilisation

The conserved tangential bias (Eq. 52) gives the sheet an inherited spin from the CMT azimuthal velocity gradient:

$$\omega_{\text{spin}} = \left. \frac{dU_{\phi}}{dr} \right|_{\text{CMT}} \cdot \frac{R_{\text{torus}}}{r_{\text{orb}}} \sim 10^{-3} \text{ rad s}^{-1}, \quad (51)$$

corresponding to $T_{\text{spin}} \approx 1.5$ h. This spin stabilises the sheet against early gravitational collapse. The spin *decreases* as the sheet moves outward (angular momentum conservation), preventing centrifugal disruption.

Phase III — Orbital capture and plastic accretion

Beyond $R_{\text{Roche}} \approx 3.6 R_{\oplus}$, tidal forces no longer fragment the material. Relative velocities between co-orbital fragments ($\delta v \approx 50\text{--}200 \text{ m s}^{-1}$, Eq. 34) are below the fragmentation threshold: collisions are sticky.

Phase	Region	Dominant process	Material state
I	$r < 2\text{--}3 R_{\oplus}$	Ballistic expansion, adiabatic cooling	Plastic, cohesive
II	$2\text{--}3 R_{\oplus}$ to orbit	Inherited spin, nascent self-gravity	Cohesive fragments
III	Orbit ($> R_{\text{Roche}}$)	Sticky collisions ($\delta v < 1 \text{ km s}^{-1}$)	Plastic accretion

L6 — Flux → PAO transition: qualitative status

The three-phase description is physically motivated and consistent with the computed orders of magnitude. The transition from a cohesive hypersonic flux to a solid-plastic body in orbit remains however described *qualitatively*. A 3D SPH simulation with coupled radiative cooling is required for quantitative validation. This is Priority 2b of the validation programme.

9 Proto-Accretion Orbital body and mass budget

The first episode places a solid-plastic body ($R_1 \approx 800\text{--}950$ km) in orbit beyond the Roche radius ($R_{\text{Roche}} \approx 3.6 R_{\oplus}$). The conserved tangential bias:

$$\mathbf{U}_{\text{ejected}} = U_r \hat{r} + U_{\phi} \hat{\phi}, \quad (52)$$

guarantees an inherited spin. The Moon builds by concentric plastic accretion, layer by layer.

$$M_{\text{Moon}} = N \cdot f_{\text{cap}} \cdot M_{\text{ej}}^{\text{max}} + \Delta M_{\text{late}}, \quad (53)$$

$N \approx 5\text{--}9$ (nominal 7 ± 2), $f_{\text{cap}} \in [0.65, 0.85]$ (nominal 0.70).

Established result

f_{cap} is a growing state variable. $\partial f_{\text{cap}} / \partial M_{\text{Moon}} > 0$: the larger the Moon, the more efficiently it captures. Combined with decreasing $g_{\text{eff}}^{(k)}$ and growing ejected masses (Prediction P20), this forms a dynamic self-reinforcing attractor.

10 Internal stratification and seismological programme

10.1 Concentric layer architecture

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k+1}, \quad k = 1, \dots, N-1. \quad (54)$$

Density contrast $\Delta\rho_{k,k+1} \approx 100\text{--}200 \text{ kg m}^{-3}$ for $\Delta\text{Mg\#} \approx 10$ (Garcia et al., 2019): potentially detectable seismic interfaces.

10.2 Two stratification scenarios

Scenario A (equal masses, $q = 1$, $N = 7$): $R_k^{(A)} = R_{\text{Moon}}(k/N)^{1/3}$, $d_k^{(A)} = R_{\text{Moon}}[1 - (k/N)^{1/3}]$.

Scenario B (growing masses, $q \approx 1.122$): both scenarios converge to a **preservation window** of approximately 200–530 km.

10.3 Preservation window: two destructive processes

From below: ilmenite cumulate overturn (Briaud et al., 2023) (episodes 1–2, > 600 km).

From above: basaltic mare volcanism to ≈ 2 Ga (Li et al., 2021; Che et al., 2021) (episodes 6–7, < 100 km). Layers 3–5 may have escaped both processes.

Falsifiable prediction

P22—Seismic interfaces in the 200–530 km window. Impedance contrasts $|R| \in [0.01, 0.04]$ detectable by:

- A single sharp interface at $d \approx 250\text{--}315$ km $\Rightarrow$ scenario $N = 2\text{--}3$;
- Several distinct interfaces at 200–530 km $\Rightarrow$ scenario $N = 5\text{--}9$: stronger confirmation.

Falsification: no phase conversion in 200–530 km window after ≥ 3 stations, ≥ 10 lunar events $M > 2.5$.

Missions: Chang’e 7 (2026), FSS, LEMS, Artemis III (2028–2029), ECLIPSE (≈ 2030).

11 Third phase transition: the progressive dynamo

$$\Delta t_{\text{dynamo}} = \underbrace{\Delta t_{\text{ejections}}}_{\approx 60 \text{ Myr}} + \underbrace{\Delta t_{\text{cooling}}}_{\approx 140 \text{ Myr}} + \underbrace{\Delta t_{\text{emergence}}}_{\approx 150 \text{ Myr}} \approx 350 \text{ Myr}. \quad (55)$$

Phase 1 (≈ 60 Myr): active Fe-Ni segregation; subadiabatic thermal gradient (Kleine et al., 2002; Yin et al., 2002). Phase 2 (≈ 140 Myr): obliquity stabilised at 23.5° ; protocrust at 4.4 Ga (Wilde et al., 2001; Valley et al., 2002). Phase 3 (≈ 150 Myr): Elsasser threshold:

$$\Lambda_{\text{Els}}^{\text{crit}} = \frac{\sigma B^2}{\rho \Omega} \approx 0.3, \quad B_{\text{crit}} \approx 6 \mu\text{T} \quad (\sigma \approx 5 \times 10^5 \text{ S m}^{-1}). \quad (56)$$

Established result

350 Myr delay with no free parameter. $\Delta t_{\text{ejections}}$: Hf-W (Kleine et al., 2002; Yin et al., 2002). $\Delta t_{\text{cooling}}$: Jack Hills zircons (4.4 Ga, Wilde et al. 2001; Valley et al. 2002). $\Delta t_{\text{emergence}}$: Tarduno et al. 2025 (4.2 Ga). Three independent observations—not an interpolation.

12 Falsifiable predictions

P1—Isotopic identity $\Delta^{17}\text{O} < 5$ ppm

Common source at each ejection. *Falsification*: if $\Delta^{17}\text{O} > 20$ ppm between deep lunar layers and the terrestrial mantle.

P3—Non-linear Hf-W chronology

Discrete episodes in W-isotope signature. *Falsification*: if the W-age distribution is continuous over 4.568–4.49 Ga.

P4—Bimodal inter-episode intervals

Early chaotic phase (closely-spaced, $< 30\,000$ yr) then Mathieu-regulated phase (widely-spaced). *Falsification*: if unimodal distribution.

P6—Increasing radial Fe/Si gradient

Fe/Si increasing with depth; SPA basin exposes Fe-richest material (episode 1). *Missions*: Chang’e-6, Artemis III.

P17—Gradual Hadean geomagnetic growth

Continuous palaeointensity growth 4.55–4.20 Ga without abrupt onset (Tarduno et al., 2025). *Falsification*: abrupt onset detected.

P22—Seismic interfaces 200–530 km (see §10)

Priority 1 of the validation programme.

13 Acknowledged limitations

L1—Spontaneous CMT organisation

Four convergent effects (Section 5) motivate the band $|\phi| < 30^\circ$, qualitatively confirmed by preliminary simulation ($N = 500, 8 T_{\text{rot}}$). Single vs. multiple coexisting structures cannot be resolved below higher-resolution simulation (Priority 2).

L2—Numerical value of f_{cap}

Physically motivated parameterisation. 3D SPH simulation is the ultimate arbiter (Prediction P12).

L3—Cohesion at small scales

$\text{Bi}_{\text{KH}} \gg 1$ holds at large and intermediate scales. Partial fragmentation at $\lambda_{\text{KH}} \ll R_{\text{torus}}$ cannot be analytically excluded (Prediction P11).

L4—Dynamo delay: ± 50 Myr

Simple exponential core-growth model. Associated uncertainty: ± 50 Myr.

L5—Application of $\sigma_{\text{res}} = \Omega f/2$ to this context

The parametric elliptic instability growth rate (Kerswell, 2002; Lacaze et al., 2004) was established for an externally forced ellipsoid in small-amplitude regime. Its application to a Maclaurin spheroid in free chaotic nutation is a physically motivated extrapolation, not a direct deduction. Level-3 hypothesis: to be validated by simulation.

L6—Flux $\rightarrow$ PAO transition: qualitative status (see §8.7)

The three-phase description is physically motivated and consistent with computed orders of magnitude, but remains qualitative. A 3D SPH simulation with coupled radiative cooling is required (Priority 2b).

L7—Crustal dichotomy: qualitative $m = 1$ asymmetry

The mechanism of crustal dichotomy formation (near-side ≈ 30 km vs. far-side ≈ 60 km, Wieczorek et al. 2013) is described qualitatively in the TPT but not yet derived analytically.

Proposed mechanism (status: qualitative). After the first episode, the PAO has

an inherited spin ($\omega_{\text{spin}} \sim 10^{-3} \text{ rad s}^{-1}$, Section 8.7) at orbital distance $\approx 3\text{--}6 R_{\oplus}$. The tidal locking time:

$$\tau_{\text{lock}} \sim \frac{\omega_{\text{spin}} a^6 Q_{\text{Moon}}}{k_2 G M_{\oplus}^2 R_{\text{Moon}}^5} \approx 10^4\text{--}10^5 \text{ yr}, \quad (57)$$

is geologically instantaneous. The PAO tidally locks while still hot ($T \approx 2000\text{--}2500 \text{ K}$).

From the second episode onward, the growing proto-lunar tide (Eq. 19) forces precession of the proto-Moon's spin axis. Combined with the thermal asymmetry between the near side (maintained warmer by terrestrial and T-Tauri fluxes) and the far side (free radiative cooling), this drives differential crystallisation:

- **Near side:** deeper lunar magma ocean, later crystallisation, thinner anorthositic crust;
- **Far side:** faster cooling, earlier crystallisation, thicker crust.

Asymmetric accretion of the final episodes ($N-1$, N), when forced precession is strongest, reinforces the nascent dichotomy. The $\approx 2 : 1$ thickness ratio may emerge from this coupling, but the coupled thermal calculation has not yet been derived analytically.

Status: **qualitative prediction** (P7— $m = 1$ asymmetry). The physical mechanism is internally consistent and requires no exogenous ingredient; it awaits confirmation or refutation by Artemis III data and dedicated thermal models.

14 Discussion

14.1 Comparison with the giant-impact model

Constraint	Giant impact	TPT
Isotopic identity	Difficult, ad hoc	Satisfied mechanically
Fe depletion	Requires differentiated Theia	Consequence of driver
Crustal dichotomy	Not predicted	Qualitative mechanism (forced precession + thermal asymmetry, L7)
Dynamo delay	Not predicted	Predicted: 350 Myr, no free parameter
Seismic predictions	None	P22 testable from 2026

14.2 Responses to principal objections

“The CMT is not demonstrated.” Not postulated: it is mode $\ell = 1, m = 1$ of the inverse cascade in a rapidly rotating oblate sphere with obliquity $> 45^\circ$. Qualitative confirmation ($E_{\text{CMT}}/E_{\text{ext}} \approx 10$). Quantitative validation is Priority 2.

“Taylor-Proudman prevents the CMT.” Columns exist but are curved by five mechanisms (Section 4). Their curvature generates the toroidal flow. The poloidal/toroidal decomposition confirms $(\boldsymbol{\Omega} \cdot \nabla)\mathbf{u}_{\text{tor}} = 0$ automatically (Frigaard & Nouar, 2017).

“ $We \approx 10^{15}$: cohesion is impossible.” Weber number applies to free flows. The CMT is a confined collective flux. Relevant criterion: $\text{Bi}_{\text{KH}} \gg 1$ (Section 7).

“The dynamo delay is arbitrary.” Three independently constrained durations (Section 11).

14.3 Sequenced validation programme

1. **Priority 1—Lunar seismology** (2026–2030): Chang’e 7, FSS, LEMS, Artemis III. Direct test of P22.
2. **Priority 2—High-resolution 3D SPH simulation:** honest assessment of $\lambda(U) = 0$ threshold, measurement of $\hat{\alpha}$, CMT validation.
3. **Priorities 3–5:** hypersonic cohesion (P11), Mathieu resonance, magma-atmosphere coupling.

15 Conclusion

This work proposes that lunar formation may be the product of a Triple Phase Transition in a fully molten proto-Earth rotating at ≈ 3.5 h, whose axis wobbles chaotically within $[40^\circ, 70^\circ]$ in its own co-rotating frame. A single driver—progressive Fe-Ni segregation—simultaneously governs three transitions: rheological, mechanical and magnetic. The result is a Moon built layer by layer, archiving in its internal structure the history of terrestrial differentiation.

The central prediction stands: one or more seismic interfaces in the 200–530 km window. If Chang’e 7 detects an interface with $|R| > 0.02$ in this window, concentric stratigraphy receives its first observational support prior to any SPH simulation. Multiple interfaces constitute stronger evidence than a single one. Absence of any interface after three stations and ten events $M > 2.5$ seriously challenges P22, and revision will be required—which the author accepts explicitly.

This work invites the community to evaluate the theory on its own terms: equations

are numbered, derivations are explicit, hypotheses are ranked, limitations are acknowledged, and predictions are falsifiable.

A Complete derivation of the λ coefficient

Solberg-Hoiland parcel-displacement formalism (Solberg, 1936; Hoiland, 1941) adapted to an oblate sphere with obliquity ε . Conservation of angular momentum (justified by $\tau_{\text{relax}}/\tau_{\text{dyn}} \approx 3 \times 10^9 \gg 1$):

$$U_{\text{parcel}}(r + \xi_r) \approx U(r)(1 - \xi_r/r), \quad (58)$$

$$U_{\text{background}}(r + \xi_r) \approx U(r)(1 - \alpha\xi_r/r). \quad (59)$$

Velocity difference: $\delta U = U(\alpha - 1)\xi_r/r$. Net force sums effective gravity, radial Coriolis and Rayleigh term. Equation (40) follows.

B Derivation of $\tau_{\text{Stokes}} \approx 200$ s

$$v_{\text{Stokes}} = \frac{2}{9} \frac{(\rho_{\text{Fe}} - \rho_{\text{sil}}) g_{\text{eff}}^{(0)} r^2}{\eta} \approx 156 \text{ m s}^{-1}. \quad (60)$$

Over $\ell \approx 30$ km: $\tau_{\text{Stokes}} \approx 192$ s. Robust over $r \in [1, 10]$ cm and $\eta \in [0.05, 1]$ Pa s.

C Adiabatic temperature profile

$T_{\text{surf}} \approx 3500$ K, gradient -0.3 K km^{-1} : central temperature ≈ 5000 K. Peridotite solidus at 135 GPa is ≈ 4300 K (Stixrude & Lithgow-Bertelloni, 2014): melting stable throughout the mantle. Confirms H0.

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